

EC2010

The Changing Economy [Macroeconomics]: Money and Monetary Policy

Summer 2022 Examination · Paper 1

Duration 1.5 hours · Section A: 4 of 6 short questions (50%) · Section B: 1 of 2 long questions (50%)

Module	EC2010 — The Changing Economy [Macroeconomics]: Money and Monetary Policy
Session	Summer 2022
Structure	Section A: six short questions, answer any four (50% of the paper between them). Section B: two long questions, answer one (50%).
This scheme covers	All six Section A questions and both Section B questions in full.
Marks convention	This paper weights each question part in "per cent" of that question's own total, rather than raw marks out of 100 — the scheme below follows the paper's own convention throughout.

Own-figure rule. If you carry an early error into a later part of a question but apply the right method consistently from that point on, you still pick up the marks for the later steps done correctly on your own figures.

How this paper relates to the other EC2010 papers. A4 (Liquidity Preference) and A5(a) (the reaction function) reappear on later papers — here A5(a) uses *general* coefficients (a, b) rather than the specific 1.5/0.5 seen from Summer 2023 onward, and A4 asks for the foundational short-run determination of interest rates rather than the LSAP-specific application seen later. A6 (Sticky Price Model) and B2(a) (evaluating interest rate effectiveness) are distinct to this sitting and don't reappear on the other papers in this set.

Section A · Question 1

A Loan vs. a Share Issue: Opposite Effects on the Money Supply · 2 × 50%

A1

100%

Explain how each of the following will affect the aggregate money supply: (In your answer you should show how each affects the assets and liabilities of the commercial bank.)

- (i) Bank of Ireland gives out a loan of €300,000 to a member of the public.
- (ii) Bank of Ireland issues shares to the value of €500,000 which it sells to the domestic (non-bank) private sector.

FULL MODEL ANSWER

(i) A €300,000 loan to a member of the public — 50%

The bank grants the loan and credits the borrower's current account with the funds — a brand-new asset (the loan) and a brand-new liability (the deposit) are created simultaneously, exactly as in the standard money-creation mechanism covered throughout this course.

Bank of Ireland balance sheet	Assets	Liabilities
Loan to borrower	+€300,000	—
Borrower's current account credited	—	+€300,000
Net effect	+€300,000	+€300,000

This new deposit is part of the money supply (M1/M3) and did not exist before the loan was made, so the aggregate money supply **increases by €300,000**.

(ii) A €500,000 share issue sold to the non-bank private sector — 50%

This is the mirror image of (i), and a common point of confusion: issuing equity is not the same as taking a deposit or extending a loan. When a buyer purchases the new shares, they pay by running down their own existing bank deposit — that deposit is destroyed, and in its place the bank records new share capital (equity).

Bank of Ireland balance sheet	Assets	Liabilities
Buyer's deposit debited to pay for the shares	No change	–€500,000 (deposits)
New share capital recorded	No change	+€500,000 (equity)
Net effect	No change	No change in total

Total liabilities are unchanged — but their composition has shifted from deposits (part of the money supply) to equity (not part of the money supply: shares have no fixed nominal value, are not a generally accepted medium of exchange, and are not included in M1/M3). The aggregate money supply therefore **decreases by €500,000**, even though the bank's balance sheet size hasn't changed at all.

Why this pairing is instructive: (i) and (ii) are deliberately symmetric — a loan creates a deposit (increasing money), while an equity sale absorbs a deposit (decreasing money). Both examples show that what matters for the money supply is specifically whether deposit money is being created or destroyed, not whether the bank's overall balance sheet is growing or shrinking. If the share buyer's account were held at a *different* bank, the mechanics run through an interbank settlement (Bank of Ireland's reserves rise, the other bank's reserves and deposits both fall), but the net economy-wide result is identical: deposits fall by €500,000.

Mark Allocation — Part (i) (50%)

Tier	Marks	What separates this tier
40-50%	Correct direction (increase) with a fully correct, correctly-labelled balance sheet entry showing both the loan and the deposit, and a clear statement that the new deposit is money.	
25-39%	Correct direction and mostly correct balance sheet entries, but the explanation is thin.	
10-24%	States the loan increases the money supply with little supporting mechanics.	
0-9%	Incorrect direction or no meaningful treatment.	

Mark Allocation — Part (ii) (50%)

Tier	Marks	What separates this tier
40-50%	Correctly identifies that the money supply <u>falls</u> because a deposit is converted into non-money equity, with a fully correct balance sheet entry.	
25-39%	Correct direction with a reasonable balance sheet entry, but the deposit-vs-equity distinction is not clearly explained.	
10-24%	States that the transaction affects the money supply with little supporting mechanics, or gets the direction wrong but shows some relevant balance sheet understanding.	
0-9%	Treats the share issue as money-neutral or as increasing the money supply, with no meaningful correction.	

Question A1 Total: 100%

Section A · Question 2

Standard & Non-Standard Monetary Policy Measures · 100%

A2

100%

Read the following extract of the ECB's Monetary Policy Decisions, Feb 3rd 2022: "The interest rate on the main refinancing operations and the interest rates on the marginal lending facility and the deposit facility will remain unchanged at 0.00%, 0.25% and -0.50% respectively. In support of its symmetric 2% inflation target and in line with its monetary policy strategy, the Governing Council expects the key ECB interest rates to remain at their present or lower levels until it sees inflation reaching 2% well ahead of the end of its projection horizon and durably for the rest of the projection horizon, and it judges that realised progress in underlying inflation is sufficiently advanced to be consistent with inflation stabilising at 2% over the medium term." Explain each of the standard and non-standard monetary policy measures mentioned in this extract.

FULL MODEL ANSWER

The standard measure: the three key ECB interest rates

What it is: the conventional, textbook monetary policy instrument — the corridor of standing facility rates (marginal lending facility 0.25%, deposit facility -0.50%) around the main refinancing operations rate (0.00%), used to steer short-term market interest rates and, through the transmission mechanism, aggregate demand and inflation. This is the tool central banks have used since long before the 2008 financial crisis.

Non-standard measure 1: Negative Interest Rate Policy (NIRP)

What it is: the deposit facility rate of -0.50% is itself a non-standard measure. Conventionally, zero was treated as an effective lower bound on policy rates — below zero, it was assumed savers would simply hold physical cash rather than pay a bank to hold their deposits. The ECB (like several other central banks post-2014) pushed policy rates below zero, betting that the costs and inconvenience of holding large amounts of physical cash would keep most deposits in the banking system even at a negative rate, allowing an additional degree of monetary easing once the conventional zero floor was reached.

Non-standard measure 2: Forward guidance

What it is: the explicit statement that rates will remain at present or lower levels until specific inflation conditions are met — inflation reaching 2% well ahead of the projection horizon and durably for the rest of the horizon, with underlying inflation also judged sufficiently advanced. This is **state-contingent forward guidance**: a pre-committed rule about the future path of policy, conditional on specific outcomes rather than a specific calendar date. Forward guidance became a standard part of the central bank toolkit only after the 2008 crisis, once conventional rate cuts were constrained by the effective lower bound — it works by shaping market expectations of future rates directly, which moves longer-term rates today even without any current change in the policy rate itself.

Why classify NIRP and forward guidance as 'non-standard': both were developed specifically because conventional interest rate policy alone became constrained once rates approached zero after 2008-09 (and again during the pandemic) — unlike the standard corridor system, which has been part of monetary policy implementation since long before the crisis. A complete answer should identify all three specific instruments (MRO/MLF/DFR — standard; the negative DFR level itself — non-standard; the explicit state-contingent commitment — non-standard) rather than treating 'the interest rates' and 'the guidance' as only two things to explain.

Mark Allocation (100%)

Tier	Marks	What separates this tier
75-100%	Correctly identifies and explains all three elements: the standard interest rate corridor, negative interest rate policy as a non-standard measure (with the effective lower bound reasoning), and state-contingent forward guidance as a non-standard measure (with the expectations-shaping reasoning).	
50-74%	Correctly identifies two of the three elements with reasonable explanation.	
25-49%	Identifies that this involves both interest rates and guidance, but doesn't correctly distinguish standard from non-standard or explain why each classification applies.	
0-24%	Minimal or largely incorrect treatment.	

Question A2 Total: 100%

Section A · Question 3

The Equation of Exchange & the Quantity Theory of Money · 100%

A3**100%**

Outline the Equation of Exchange (Fisher, 1911) and explain how this can be transformed into a theory of the relationship between Money and Prices.

FULL MODEL ANSWER**Outlining the Equation of Exchange**

Irving Fisher's 1911 formulation is the identity $MV = PT$, where M is the stock of money, V is the (transactions) velocity of circulation, P is the price level, and T is the total volume of transactions conducted using money over a given period. A more modern version replaces T with real output/income (Y), giving the more familiar $MV = PY$ used elsewhere in this course.

Symbol	Meaning
M	The stock of money in circulation
V	Velocity of circulation — the average number of times a unit of money is used in a transaction over the period
P	The general price level
T (or Y)	The total volume of transactions (Fisher's original) or real output (the modern version)

Why it's an identity: $MV = PT$ holds definitionally — total spending in the economy (the money stock times how often each unit is spent) must equal the total value of transactions conducted (the price of each transaction times the number of transactions), since every euro spent is a euro received. This is true by construction, for any values of M , V , P and T , and carries no economic content on its own.

Transforming the identity into a theory: the Quantity Theory of Money

The identity becomes a substantive **theory** of the relationship between money and prices once two additional (non-definitional) assumptions are imposed:

- **Assumption 1 — V is stable (at least in the medium run).** Fisher argued velocity is determined by payment habits, institutions, and technology (how often people are paid, how quickly cheques clear, how commonly credit is used) — factors that change only slowly, and independently of M , P , or T . This means V can be treated as approximately constant for the purposes of medium-run analysis.
- **Assumption 2 — T (or Y) is determined independently by real factors.** In the classical view, the volume of transactions/real output is determined by the economy's productive capacity (labour, capital, technology) — the 'real side' of the economy — and is essentially unaffected by the quantity of money in circulation, at least in the long run (monetary neutrality).

The resulting theory: with V and T (or Y) both effectively fixed independently of M , the identity $MV=PT$ rearranges to $P = MV/T$: the price level is proportional to the money stock. A given percentage change in M translates into an equal percentage change in P , with causation running from money to prices — the core proposition of the **Quantity Theory of Money**, and the classical origin of the idea that inflation is, at root, a monetary phenomenon.

What this adds beyond the identity: the identity alone tells us nothing about causation or magnitude — it could equally be read as 'P and T determine what MV must have been'. It is only the added behavioural assumptions about V and T that turn $MV=PT$ into a predictive, testable theory: specifically, that money growth causes proportional price inflation, holding V and T fixed.

Where the theory can break down (and why this matters for later questions on this paper): if V is not actually stable — for example, if a rise in M is absorbed by a fall in V rather than a rise in P — then the same identity $MV=PT$ still holds exactly, but the quantity-theory *prediction* (that M and P move proportionally) fails. This is precisely the issue explored in Q. B1(b) on this paper, regarding the weak link between QE and inflation in the Euro Area.

Mark Allocation (100%)

Tier	Marks	What separates this tier
75-100%	States $MV=PT$ (or $MV=PY$) correctly, defines every term, explains why it is an identity, and correctly identifies both assumptions (stable V , independently-determined T/Y) needed to transform it into the Quantity Theory of Money, with the resulting proportionality result $P=MV/T$.	
50-74%	States the equation correctly and defines terms, but the identity-vs-theory transformation (i.e. what assumptions are needed and why) is only partially explained.	
25-49%	States the equation with minor errors in definitions or no real discussion of the transformation into a theory.	
0-24%	Incorrect or missing equation.	

Question A3 Total: 100%

Section A · Question 4

The Theory of Liquidity Preference · 100%

A4

100%

Use the Theory of Liquidity Preference to explain how interest rates are determined in the short run. You should use a diagram to support your answer.

FULL MODEL ANSWER

Setting up the model

Keynes's **Theory of Liquidity Preference** explains the interest rate as the price that equates the supply of and demand for money (rather than the price that equates saving and investment, as in the classical loanable funds theory). This is a genuinely short-run theory: it takes the money supply as fixed by the central bank and asks what interest rate makes people willing to hold exactly that quantity of money.

Interest Rate Determination in the Theory of Liquidity Preference

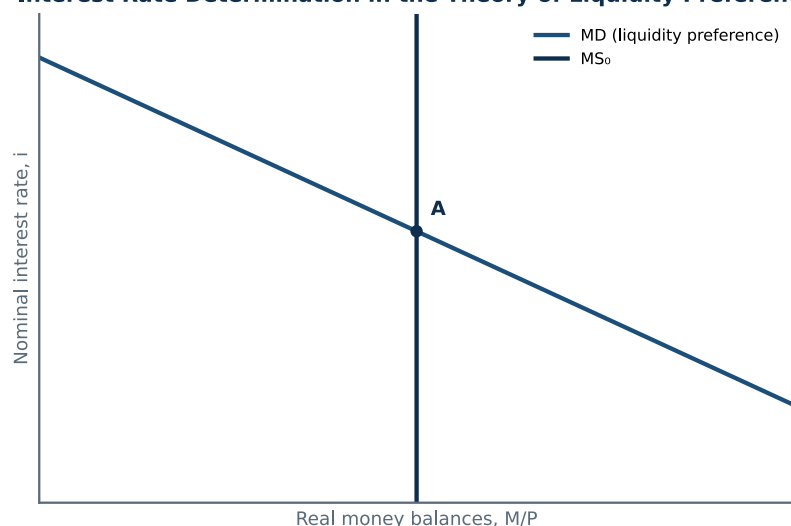


Fig. 1 — Interest rate determination: the intersection of money demand and the (fixed) money supply

Money demand (liquidity preference)

Keynes identified three motives for holding money rather than interest-bearing assets:

- **Transactions motive:** money is held to bridge the gap between receiving income and making planned purchases — depends mainly on the level of income/output, not the interest rate.
- **Precautionary motive:** money is held as a buffer against unexpected expenses or opportunities — also mainly income-dependent.
- **Speculative motive:** money is held as an alternative to bonds when the interest rate is expected to rise (since a bond's price would then fall) — this is the motive that makes money demand interest-rate dependent. At a higher interest rate, the opportunity cost of holding non-interest-bearing money is higher, so people economise on money holdings and hold more bonds; at a lower rate, they're more willing to hold idle money.

Result: aggregate money demand (MD) slopes downward in interest-rate/money space, primarily reflecting the speculative motive's sensitivity to the interest rate.

Money supply

Vertical MS curve: the central bank sets the nominal money supply directly (through the tools examined in Q. B1(a) — open market operations, or via the interest-rate channel and its effect on bank lending) largely independently of the interest rate itself, so MS is drawn as a vertical line at the quantity the central bank has chosen to supply.

Short-run equilibrium and the adjustment mechanism

Equilibrium: the interest rate adjusts until money demand exactly equals the fixed money supply — the intersection of MD and MS.

How the adjustment actually happens: if the interest rate starts above equilibrium, money demand falls short of the available supply — people are holding more bonds and less money than they want at that rate, so they buy bonds with their excess money, bidding bond prices up and yields (interest rates) down until equilibrium is restored. If the rate starts below equilibrium, the reverse happens: people sell bonds to hold more money, pushing bond prices down and yields up. This portfolio-rebalancing process is what actually determines the interest rate in the short run, and is the same underlying mechanism used to analyse large-scale asset purchases in the equivalent question on the Summer 2023 paper — there, an outward shift in MS (from central bank asset purchases) works through exactly this mechanism to lower the equilibrium rate.

Why 'short run' matters: this theory takes the price level and real output as roughly fixed over the period being analysed, and explains only the interest rate. It is not a theory of how output or prices are determined — those come from combining this with the other models in the course (IS-TR, AD-AS).

Mark Allocation (100%)

Tier	Marks	What separates this tier
75-100%	Correct MD/MS diagram, explains all three motives for holding money (or at minimum correctly identifies the speculative motive as the source of MD's interest-rate sensitivity), correctly draws MS as vertical, and explains the bond-price adjustment mechanism that restores equilibrium.	
50-74%	Correct diagram and basic explanation of MD/MS, but the adjustment mechanism (how the rate actually moves to equilibrium) is missing or unclear.	
25-49%	Some correct elements but the diagram or the MD/MS roles are confused.	
0-24%	Minimal or largely incorrect treatment.	

Question A4 Total: 100%

Section A · Question 5

The General Reaction Function & Central Bank Stance · 60% + 2×20%

A5(a)**60%**

The following equation represents the reaction function of a Central Bank:

$$i = i^n + a(\pi - \pi^T) + b(Y - Y^T)/Y^T$$

where i^n = Natural (Neutral) Rate of Interest, π^T = Inflation Target, Y^T = Trend Output. Using this reaction function, explain how the Central Bank changes interest rates.

A5(b)**40%**

How might you expect the reaction function to differ for:

- (i) A hawkish Central Bank;
- (ii) A dovish Central Bank? Briefly explain your answer.

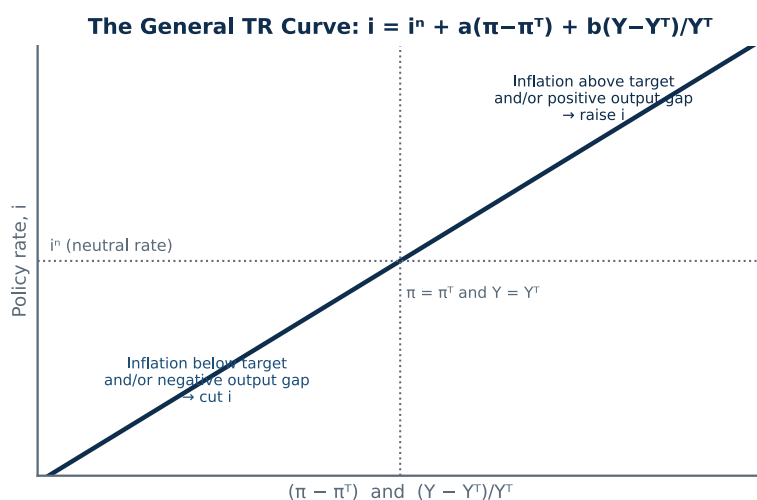
FULL MODEL ANSWER**(a) Interpreting the general reaction function — 60%**

Fig. 1 — The TR curve implied by this reaction function; its slope is governed by a and b

Structure: the Central Bank sets its policy rate (i) equal to the neutral rate (i^n) plus two adjustment terms, each scaled by a coefficient (a and b) that determines how aggressively the bank responds to each gap.

The inflation-gap coefficient, a : if inflation is above target, this term is positive and the bank raises rates above neutral; if below target, it cuts rates. For this response to be stabilising in real terms, **a must exceed 1** (the Taylor principle) — since the real rate is approximately $i - \pi$, a coefficient below 1 would mean a rise in inflation actually lowers the real rate, which would be destabilising rather than corrective.

The output-gap coefficient, b : if output is above trend, this term is positive and the bank raises rates, anticipating inflationary pressure; if below trend, it cuts rates to support demand. Unlike a , there's no equivalent lower-bound restriction on b for stability — b simply governs how much weight the bank places on output stabilisation relative to inflation stabilisation.

Baseline case: when $\pi=\pi^T$ and $Y=Y^T$, both terms vanish and $i=i^n$ regardless of the values of a and b — the bank holds the neutral rate exactly at target.

(b) Hawkish vs. dovish reaction functions — 2 × 20%

Central Bank stance	Expected coefficients	Interpretation
(i) Hawkish	Higher a (further above 1); possibly lower b	Prioritises inflation control above output/employment stabilisation; raises rates more aggressively per unit of inflation gap, and is willing to accept a larger output cost to bring inflation back to target quickly; keeps rates 'higher for longer' when inflation is elevated.
(ii) Dovish	Lower a (closer to, but still above, 1); possibly higher b	Still satisfies the Taylor principle ($a>1$) but responds less aggressively to a given inflation gap, places relatively more weight on supporting output/employment, and is more tolerant of a temporary inflation overshoot if it means avoiding a sharp slowdown; keeps rates lower for longer, all else equal.

Why this matters in practice: the same current data (say, inflation 1pp above target with output at trend) will generate a materially different policy rate depending on whether the Governing Council leans hawkish or dovish at that particular meeting — the reaction function's coefficients aren't just an abstract equation, they summarise a genuine, observable difference in central bank behaviour and communication style.

Keep b 's role in mind too: a strong answer for part (b) doesn't only discuss a — a hawkish bank plausibly also has a *lower* b (caring relatively less about the output gap), since a bank that weights output stabilisation heavily is, by construction, more dovish on inflation trade-offs. The two coefficients should be discussed as a pair, not just a in isolation.

Mark Allocation — Part (a) (60%)

Tier	Marks	What separates this tier
45-60%	Correctly explains the role of both coefficients (a and b), explicitly states and explains the Taylor principle ($a>1$) for stability, and states the baseline case.	
30-44%	Explains that the bank responds to each gap via a and b but misses the Taylor principle or the baseline case.	
15-29%	General description without engaging with the role of the coefficients.	
0-14%	Minimal or incorrect interpretation.	

Mark Allocation — Part (b) (2 × 20%)

Tier	Marks	What separates this tier
16-20% each	Correctly identifies the expected direction of both a and b for the given stance, with clear economic reasoning for why.	
8-15% each	Correctly identifies the direction of a but doesn't discuss b , or reasoning is thin.	
0-7% each	Minimal or incorrect treatment.	

Section A · Question 6

Deriving the SRAS Curve: The Sticky Price Model · 100%

A6**100%**

Derive the short run Aggregate Supply Curve using the Sticky Price Model.

FULL MODEL ANSWER**Setting up the model**

The Sticky Price Model explains the upward-sloping short-run aggregate supply (SRAS) curve from firm-level price-setting behaviour under imperfect information/costly price adjustment, rather than assuming it. Each firm i wants to set a price p_i based on two things: the overall price level in the economy (P) and the state of demand relative to the natural rate of output ($(Y - \tilde{Y})/\tilde{Y}$):

$$p_i = P + a(Y - \tilde{Y})$$

where $a > 0$: a firm wants to charge a relatively higher price than the average when aggregate output/demand is above its natural rate (strong demand justifies a higher relative price), and a relatively lower price when demand is weak.

Step 1 — Two types of firms

- **Flexible-price firms (a fraction $1-s$ of firms):** set their price after observing the current state of the economy, so they set $p_{\text{flex}} = P + a(Y - \tilde{Y})$ using the actual current P and Y .
- **Sticky-price firms (a fraction s of firms):** must set their price in advance, before observing the current P and Y — they set their price based on what they expect P and Y to be: $p_{\text{sticky}} = P^e + aE[Y - \tilde{Y}]$. Since firms have no systematic reason to expect output to differ from its natural rate on average, $E[Y - \tilde{Y}] = 0$, so $p_{\text{sticky}} = P^e$.

Step 2 — Aggregating to the overall price level

The overall price level is the weighted average of both groups' prices:

$$P = s \cdot P^e + (1-s) \cdot [P + a(Y - \tilde{Y})]$$

Solving for P (subtract $(1-s)P$ from both sides):

$$P = P^e + [(1-s)/s] \cdot a \cdot (Y - \tilde{Y})$$

Step 3 — Rearranging to the standard SRAS form

Rearranging for Y gives the short-run aggregate supply curve:

$$Y = \tilde{Y} + \alpha(P - P^e), \quad \text{where } \alpha = s / [(1-s)a]$$

The Short-Run Aggregate Supply Curve (Sticky Price Model)

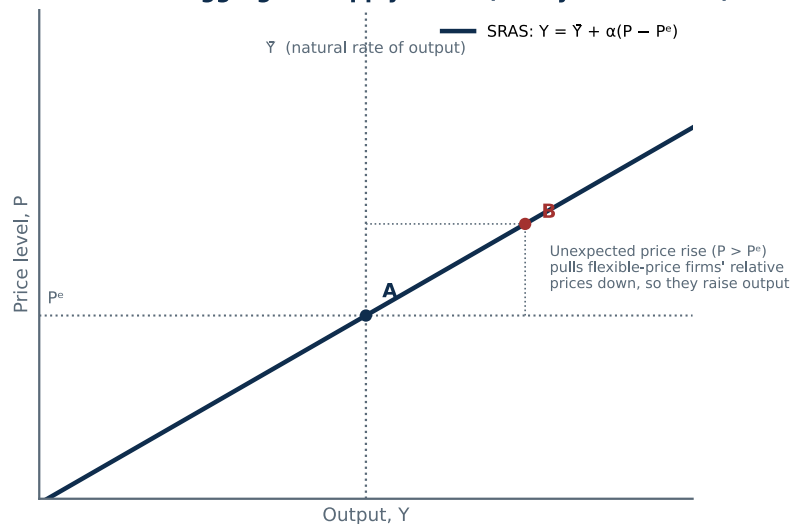


Fig. 1 — The short-run AS curve: output rises above the natural rate whenever the actual price level exceeds what firms expected

Interpreting the result

Why it slopes upward: when the actual price level (P) turns out higher than firms expected (P^e) — an unanticipated rise in demand or prices — the flexible-price firms raise their prices in response, but the sticky-price firms are stuck at their pre-set (lower, relatively speaking) price. Because sticky firms' prices are now relatively low, demand for their output rises, and total output (Y) rises above $\bar{Y}$. A price surprise therefore raises output: the essence of the short-run AS curve's upward slope.

The role of s (the sticky-price share): when s is large (many firms have sticky prices), the curve is flatter — a given price surprise moves output by more, because a bigger share of firms are 'stuck' and absorb the demand shift through quantity rather than price. When s is small (most firms are flexible), the curve is steeper — closer to the vertical long-run AS curve, since most firms adjust price immediately and little of the demand shift shows up as an output change.

The vertical long-run case: in the long run, $P^e = P$ (expectations catch up with reality, since firms eventually observe and adjust), so $P - P^e = 0$ and $Y = \bar{Y}$ regardless of the price level — the long-run AS curve is vertical, exactly as required for monetary neutrality in the long run.

Exam tip: 'derive' means you're expected to show the algebra (the p_i equation, the weighted average, the rearrangement), not just state the final $Y = \bar{Y} + \alpha(P - P^e)$ result. Full marks require the intermediate steps shown above.

Mark Allocation (100%)

Tier	Marks	What separates this tier
75-100%	Full correct derivation shown step by step (individual firm pricing rule, flexible vs. sticky firms, aggregation, rearrangement to $Y = \bar{Y} + \alpha(P - P^e)$), with correct interpretation of the slope and the role of s .	
50-74%	States the correct final SRAS equation and a broadly correct interpretation, but the algebraic derivation steps are incomplete or skipped.	
25-49%	States that SRAS is upward-sloping with a general intuition (price surprises raise output) but no real derivation.	
0-24%	Minimal or largely incorrect treatment.	

Section B: Long Questions

Answer 1 of 2 · this scheme covers both in full

Section B questions are broader and typically integrate several ideas together.

Study-Uni

Section B · Question 1(a)

Constraints Banks Face on Their Own Money Creation · 3 × 20%

B1(a)**60%**

Can banks create as much money as they would like? Examine the constraints that banks themselves face that limit their lending and the volume of money created. In your answer, you should show the relevant transactions on the balance sheets of the commercial banks to support your answer.

FULL MODEL ANSWER**No — banks face several binding constraints on their own**

While banks do create money "out of nothing" the moment they extend a loan (see Q. A1), this does not mean lending is unlimited. Three distinct, self-imposed (or regulatorily-imposed but internally binding) constraints limit how much money banks can actually create.

Constraint 1: Capital adequacy — 20%

Mechanism: regulators require banks to hold a minimum amount of capital (shareholders' own funds) relative to their risk-weighted assets (loans, weighted by riskiness). Every new loan increases risk-weighted assets and therefore uses up some of the bank's spare capital headroom; once a bank's capital ratio approaches the regulatory minimum, it cannot make further loans without raising additional capital.

Illustrative example	Value
Bank's capital base	€12,000,000
Minimum capital adequacy ratio	8%
Maximum risk-weighted lending supportable	€150,000,000

Bank balance sheet at the capital ceiling	Assets	Liabilities
Loans (at risk-weighted maximum)	€150,000,000	—
Deposits created by that lending	—	€138,000,000
Capital (equity)	—	€12,000,000
Total	€150,000,000	€150,000,000

Even with willing, creditworthy borrowers queuing up, this bank cannot lend beyond €150m of risk-weighted assets without raising more capital — a hard ceiling that has nothing to do with deposit funding or borrower demand.

Constraint 2: Liquidity risk — 20%

Mechanism: regulatory liquidity requirements (the Liquidity Coverage Ratio and Net Stable Funding Ratio) require a bank to hold sufficient high-quality liquid assets and stable funding relative to its balance sheet. Every new loan, if not matched by correspondingly stable funding, erodes this liquidity buffer. A bank near its liquidity limits must slow new lending even where capital is not the binding constraint, since further lending would leave it dangerously exposed to a funding squeeze or deposit outflow.

Balance sheet expression: a loan drawn down and then spent typically results in reserves flowing out to another bank (see Q. A1/B1(a) on other papers) — a bank funding a large volume of new lending primarily with short-term, flighty deposits (rather than stable retail deposits or longer-term wholesale funding) faces a much tighter liquidity constraint on how much it can safely lend than one with a stickier funding base.

Constraint 3: Credit risk assessment — 20%

Mechanism: even absent any capital or liquidity constraint, a profit-maximising, prudentially-managed bank will not lend to every applicant — each additional loan carries a probability of default that must be weighed against the interest earned. Beyond some point, expanding into progressively riskier borrowers raises expected losses faster than expected interest income, making further lending unprofitable (and, past a certain point, threatening the bank's own solvency) even if capital and liquidity headroom remain.

Commercial bank balance sheet impact of a declined risky loan application	Assets	Liabilities
Loan application assessed as too risky at the offered rate: declined	No change (loan not made)	No change (no deposit created)
Compare: if the loan had been made	Loans +X (foregone)	Deposits +X (foregone)

Net effect: credit risk assessment acts as a self-imposed brake on money creation, independent of the capital and liquidity constraints above — even a bank with ample capital and liquidity headroom will not (and, to remain solvent, should not) lend indiscriminately.

The overall answer: No, banks cannot create as much money as they like. Capital adequacy, liquidity risk, and credit risk assessment are three independent, potentially simultaneously-binding constraints — a bank might be constrained by any one of them even when the other two have headroom, so a complete answer should treat all three as separate limits rather than a single generic 'banks are careful' statement.

Mark Allocation (60%, 20% per constraint)

Tier	Marks	What separates this tier
16-20% per constraint	Correct mechanism, with a correct and clearly-labelled balance sheet illustration, and a clear statement of how this specifically limits money creation.	
8-15% per constraint	Correct mechanism identified with a reasonable balance sheet treatment, but the link to money creation is not fully drawn out.	
0-7% per constraint	Minimal or incorrect treatment of that constraint.	

Section B · Question 1(b)

Why QE's Effects on Money and Prices Were Weaker Than Expected · 40%

B1(b)**40%**

Quantitative easing (Large Scale Asset Purchase Programmes) have been used in the Euro Area between 2015 and 2020 with the aim of increasing price inflation in line with its objective. However, there appears in the short/medium term a weakening of the link between the monetary base and the money supply on the one hand, and between the money supply and the general level of prices on the other. Use the economic theory you studied in EC2010 and your understanding of money creation to suggest reasons why this is the case.

FULL MODEL ANSWER

Two distinct links are asked about here, and they weaken for different reasons. A strong answer keeps them separate rather than offering one generic explanation for both.

Link 1: Why the monetary base → money supply link weakened

The money multiplier mechanism: in the textbook model, an increase in the monetary base (reserves) supports a multiple expansion of broad money (deposits) as banks lend out a fraction of new reserves, those loans become new deposits at other banks, and the process repeats — the multiplier depends on banks actually choosing to lend out their excess reserves.

Why the multiplier broke down post-2015: banks, facing tighter post-crisis capital and liquidity regulation (see Q. B1(a)), chose to hold a much larger share of the new reserves created by APP as precautionary liquid assets rather than lending them out — especially as near-zero and negative policy rates (Q. A2) compressed lending margins and made additional risk-taking less attractive. Weak loan demand from a economy still recovering from the sovereign debt crisis reinforced this: even willing lenders found relatively few willing, creditworthy borrowers wanting to expand their own borrowing.

Result: the monetary base expanded dramatically (reserves created directly by the ECB's asset purchases), but broad money (M3, the actual money supply relevant for spending) grew much more modestly — the money multiplier fell well short of its textbook value, weakening the base-to-money-supply link.

Link 2: Why the money supply → prices link weakened

The Quantity Theory's key assumption failed: recall from Q. A3 that $MV=PT$ becomes a theory of prices only once velocity (V) is assumed stable. Even for the portion of new money that did reach broad money and household/firm deposits, velocity fell sharply over this period — people and firms, facing weak growth prospects and elevated uncertainty, chose to hold the additional money rather than spend it at the previous rate.

Why velocity fell: in a low-growth, low-confidence environment (the aftermath of the sovereign debt crisis, weak investment appetite), the opportunity cost of holding money was itself very low (given near-zero/negative interest rates), so there was little penalty to simply holding additional money balances rather than spending them — a form of increased money demand (precautionary and speculative motives, see Q. A4) that absorbed the extra money without translating into extra spending.

Result: even the broad money that was created didn't translate proportionally into higher spending and prices, because falling velocity offset rising M in the identity $MV=PY$ — consistent with the identity holding exactly, but the quantity-theory *prediction* failing in the short-to-medium run.

Bringing the two links together

The overall chain from “Central Bank buys assets” to “prices rise” has two separate hand-offs — base to broad money, and broad money to prices — and both weakened simultaneously over 2015-2020: banks sat on reserves instead of lending (weakening the first link), and the money that was created was increasingly held rather than spent (weakening the second). Either weak link alone would have muted APP's effect on inflation; together, they largely explain why such a large expansion of the monetary base coincided with such subdued inflation for most of this period.

What a strong answer needs: explicitly separate the two links asked about, use the money-multiplier concept for the first and the $MV=PY/\text{velocity}$ concept for the second, and explain the specific post-crisis conditions (regulation, weak demand, low rates, uncertainty) driving each — not just assert that 'QE didn't work as expected'.

Mark Allocation (40%)

Tier	Marks	What separates this tier
30-40%	Correctly and separately addresses both weak links: the base-to-broad-money multiplier (weak lending appetite, regulatory constraints, weak loan demand) and the money-to-prices link (falling velocity, low opportunity cost of holding money, weak confidence), with correct use of the money-creation mechanics and the $MV=PY$ identity.	
20-29%	Addresses both links but the explanation of one is significantly thinner than the other, or the underlying theory is only loosely applied.	
10-19%	General discussion of 'QE didn't cause much inflation' without distinguishing the two specific weak links asked about.	
0-9%	Minimal or off-topic answer.	

Question B1 Total: 100% (60% + 40%)

Section B · Question 2(a)

Evaluating the Effectiveness of Interest Rates · 60%

B2(a)**60%**

In response to a downturn in the economy, interest rates have traditionally been used as the monetary policy instrument to increase production, employment and income. Evaluate the effectiveness of interest rates at increasing output, employment and income. In your answer you should draw on the economic models and the module readings in EC2010 to support your answer.

FULL MODEL ANSWER

A genuine evaluation needs both the case for interest rate policy's effectiveness (the standard transmission mechanism) and its limitations — particularly ones exposed by the post-2008 experience that motivated the shift toward non-standard tools examined elsewhere in this course.

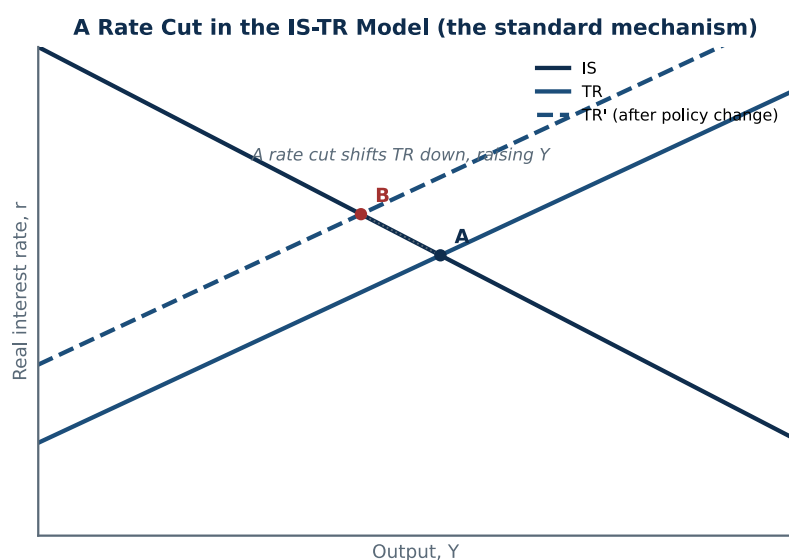
The case that interest rates are effective

Fig. 1 — The standard mechanism: a rate cut shifts TR down, raising equilibrium output

The standard transmission mechanism (IS-TR): a rate cut lowers the real interest rate, stimulating interest-sensitive spending (investment, durable consumption) and moving the economy up the IS curve to higher output — the textbook mechanism underlying the downward-sloping IS curve used throughout this course.

Historical track record: conventional interest rate policy is credited with helping to moderate ordinary business-cycle downturns for decades before 2008, and remains the default, fastest-acting instrument available — a rate decision can be implemented within days, unlike fiscal policy which typically requires legislative action.

Multiple reinforcing channels: as discussed for a rate rise in the equivalent question on the Summer 2023 paper, a rate cut works through several channels simultaneously (direct borrowing cost, bank lending/credit supply, exchange rate depreciation supporting net exports, asset price/wealth effects, and expectations) — when all channels are operating normally, this gives interest rate policy considerable reach.

The case that effectiveness is limited, especially in a deep downturn

The effective lower bound: interest rates cannot be cut indefinitely — even with negative rates (Q. A2), there is a practical floor beyond which further cuts become ineffective or counterproductive (bank profitability is squeezed, and savers may shift to physical cash). In a sufficiently deep downturn, the interest rate needed to fully restore output may lie below this floor, leaving conventional policy unable to do the whole job — precisely the situation that motivated large-scale asset purchases and other non-standard tools (Q. A4, B1(b)) from 2015 onward.

Weak transmission when banks themselves are impaired: if banks are simultaneously repairing damaged balance sheets (as after 2008), they may not pass rate cuts through to lending fully — widening their own lending margins to rebuild capital instead, blunting the bank-lending channel specifically even while headline policy rates fall.

Weak credit demand independent of price ('pushing on a string'): in a sufficiently severe downturn, households and firms may be focused on paying down existing debt (deleveraging) rather than taking on new borrowing at any interest rate — a phenomenon associated with 'balance sheet recessions' (Koo). If desired borrowing is near zero regardless of price, cutting the price of credit further does little to raise actual borrowing and spending.

Long, variable, and uncertain lags: even where the mechanism works, the 12-24 month lag between a rate change and its peak effect on output means interest rate policy responds only slowly to an unfolding downturn, and the eventual magnitude of the effect on output/employment is itself uncertain, especially for unusually large moves during unusual crises.

Evidence from 2008-2015: despite policy rates falling to near zero across major economies within a year or two of the 2008 crisis, output and employment recovery remained slow for years afterward — strongly suggesting conventional interest rate policy alone was insufficient to restore output quickly in a sufficiently deep, credit-driven downturn, which is exactly why central banks turned to LSAPs and other non-standard measures as a supplement, not a replacement.

Overall evaluation

Interest rates remain a genuinely effective tool for ordinary cyclical downturns, where the required rate cut is well within the space available above the effective lower bound and the banking system/private sector balance sheets are healthy. Their effectiveness is much more limited in deep, balance-sheet-driven downturns — exactly the kind that pushes rates toward the effective lower bound and where credit demand and bank willingness to lend are themselves impaired — which is precisely why the Euro Area (and other major economies) supplemented conventional rate policy with LSAPs, forward guidance, and other non-standard tools after 2008 rather than relying on interest rates alone.

What separates a strong essay here: don't just list channels and limitations — explicitly connect the limitations back to *why* non-standard tools became necessary (i.e. this question sets up the reasoning behind Q. A2, A4, B1(b) on this same paper), and reach a genuine evaluative verdict rather than a neutral list of pros and cons.

Mark Allocation (60%)

Tier	Marks	What separates this tier
45-60%	Presents a genuine evaluation: the standard IS-TR transmission mechanism and multiple channels for effectiveness, AND at least three distinct limitations (effective lower bound, impaired bank transmission, weak credit demand/balance sheet recession, transmission lags), reaching a clear, well-supported overall verdict linking to why non-standard tools became necessary.	
30-44%	Covers both the case for and against effectiveness but with less depth on the limitations, or without a clear overall verdict.	
15-29%	Describes the standard transmission mechanism only, with little or no discussion of limitations — a one-sided description rather than an evaluation.	
0-14%	Minimal or largely incorrect treatment.	

Section B · Question 2(b)

How Should the ECB Respond to the February 2022 Inflation Figures? · 40%

B2(b)**40%**

In February 2022, Euro area annual inflation was expected to be 5.8%, up from 5.1% in January according to a flash estimate from Eurostat, the statistical office of the European Union. How do you think that the ECB should respond to inflation figures such as these? Explain your answer.

FULL MODEL ANSWER

This is an open policy-judgement question, set at a genuinely uncertain moment: inflation had been rising steadily through 2021 and into early 2022, but the ECB (see Q. A2's extract, dated the day before this release) was still committed to holding rates at their present or lower levels pending durable evidence of inflation at target. A strong answer weighs the case for immediate action against the case for continued caution, and reaches a reasoned recommendation.

The case for responding immediately (tightening now)

The trend is accelerating, not just elevated: the rise from 5.1% to 5.8% in a single month, on top of many months already well above the 2% target, suggests inflation momentum is building, not stabilising — waiting for 'durable' evidence of a return to target, as the ECB's own forward guidance specified (Q. A2), risks falling further behind an accelerating trend.

Risk of de-anchoring inflation expectations: the longer actual inflation runs well above target, the greater the risk that households and firms begin to expect persistently higher inflation and build it into wage demands and price-setting — once expectations de-anchor, restoring credibility typically requires a much larger and more costly tightening than acting earlier would have.

Negative real rates were historically extreme: with the deposit facility at -0.50% (Q. A2) and inflation at 5.8%, the real policy rate was deeply negative (roughly -6%) — an extraordinarily accommodative stance to maintain given how far inflation already exceeded target.

The case for continued caution (holding rates, only beginning to normalise gradually)

Much of the inflation was plausibly supply-driven and outside monetary policy's effective reach: energy prices and pandemic-related supply chain disruption were major contributors to the 2021-22 inflation surge (see Q. A6's energy shock analysis) — interest rate policy has little direct effect on global energy prices or supply chain bottlenecks, and tightening in response to a supply shock imposes an output cost without addressing its root cause (the AD/AS policy trade-off examined elsewhere in this course).

Premature tightening risks damaging a still-incomplete recovery: the Euro Area economy was still recovering from the pandemic shock in early 2022; raising rates aggressively into an already-fragile recovery risked choking off growth just as it was gaining traction, especially given the long lags before any tightening would even affect inflation (Q. B2(a)'s discussion of transmission lags).

Genuine uncertainty about persistence: distinguishing a transitory, reopening-driven inflation spike from a persistent, demand-driven one is genuinely difficult in real time (see the equivalent 'rising vs. falling inflation' question on the Summer 2023 paper) — over-reacting to what turns out to be transitory is itself a policy error.

A reasoned recommendation

The most defensible response at this specific juncture is a **measured shift toward normalisation, without an immediate large rate hike**: begin visibly winding down the scale of asset purchases (a first, less disruptive step than raising rates directly), sharpen forward guidance to signal that rate increases are now firmly on the table pending further data (moving away from the very open-ended 2 February wording), and closely monitor underlying/core inflation and wage growth for signs that the shock is broadening beyond energy into a more persistent, domestically-driven dynamic. This preserves flexibility while responding to the genuine acceleration in the data, without committing to the kind of aggressive immediate tightening that could prove to be a costly overreaction if much of the surge does turn out to be supply-driven and temporary.

What actually happened (for study reference — postdates the exam, not required for full marks): the ECB held rates unchanged through Q1/Q2 2022, ended net asset purchases under APP in July 2022, and then began raising rates for the first time in over a decade with a 50bp hike on 27 July 2022 — a path broadly consistent with the 'normalise first, then hike' recommendation above, though the subsequent pace of tightening through 2022 (accelerated further by the war in Ukraine's additional energy shock from late February 2022, just after this release) turned out to be considerably faster and larger than almost anyone anticipated at the time of this question.

Mark Allocation (40%)

Tier	Marks	What separates this tier
30-40%	Presents a genuine, balanced evaluation of both the case for immediate tightening and the case for continued caution, engaging with the transitory-vs-persistent question and the transmission-lag issue, and reaches a clear, well-reasoned recommendation.	
20-29%	Presents one side of the argument reasonably well but doesn't seriously engage with the opposing case, or the recommendation isn't clearly justified.	
10-19%	General discussion of 'the ECB should watch inflation' without a substantive case either way.	
0-9%	Minimal or off-topic answer.	

Marking note: since this is a judgement question, a well-reasoned answer recommending immediate tightening should score just as highly as one recommending continued caution, provided the case is argued with genuine engagement with both sides. Penalise a one-sided answer or an unsupported assertion, not a particular directional conclusion.

Question B2 Total: 100% (60% + 40%)

